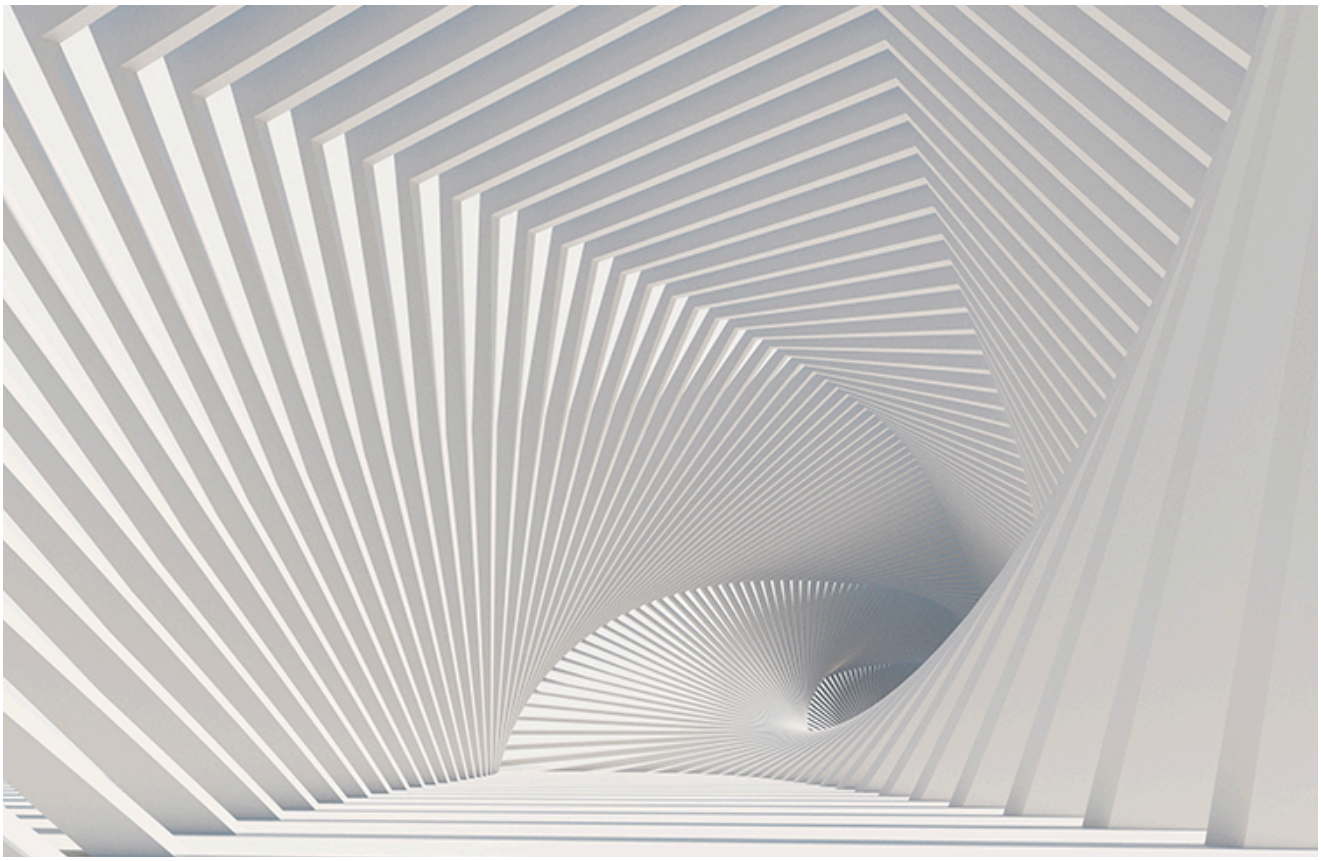
LEAGUE OF LEGENDS WIN PREDICTOR



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LEAGUE OF LEGENDS

Impact of Model

Individual: Due to League of Legends' chaotic nature, a win-probability model would reframe the game into something more measurable and predictable. This would make it so that when watching esports, the viewers start understanding the momentum of a team more objectively; tournament casters can rely less on the intuition of previous games and more on data which gives a percentage based on in-game statistics. This would also allow for players to become aware of what could be possible hidden advantages which boost their own chance of winning more games. This kind of model in terms of e-sports would shift the culture from assumptions to more reliable data based on tempo. Gold curves, exp advantages and more also elevate the analytical side of the game, as when 'Worlds' comes around, viewers flock to *Caedrel* to understand the game more deeply, as the tools for them to analyse it themselves do not exist. Players would also learn about what actually matters in the early game, gaining clarity on how small advantages can eventually lead to a 100% winnable game, although this may compromise some aspects of the individual's game, as they may become overconfident when the model favours them and eventually lead to throwing the game, creating a mindset where players may give up too early in a game. This would contribute to global e-sports literacy as this model pushes towards more analysis, more informed debates, and more sophistication on the fan culture, boosting popularity and engagement.

Society: Due to League of Legends demographic being mainly 16-30 year olds, a model on the analytical side of league could teach many young players to interpret probabilities, statistics, uncertainty, and performance metrics, pushing society toward greater data literacy—which is key in our society as more things become digitalised—just as how statistical analytics reshaped how people thought about sports such as soccer, basketball, etc. increasing analytics on Esports trains a young generation to understand key concepts through numbers and models, taking them to track on the way our current societies are progressing. This model would contribute to how society views and interacts with AI. But through this some ethical challenges may be raised; this system of prediction could increase the influence of betting behaviour on League of Legends players, turning League of Legends into a similar state Pokemon is currently in with people’s monetisation of game features

Environmental: Everything on the web runs on a server which consumes electricity, this means that is the model is often updating or processing large amounts of data, the energy demand of the overall model would increase. If this gets to scale this can contribute towards the global carbon footprint of data centres, which are responsible for a massive share of worldwide electricity consumption, for example ChatGPT consumes around 0.3 to 2.9 watt-hours of electricity, which is roughly around 10 times the amount of energy consumed compared to a google search. When thinking of the model at scale, developers will retrain models, upgrade hardware, and store increasing amounts of data accelerating e-waste cycles, as old components are replaced with newer technological advancements, which may lead to the increased dumping of old electrical compartments.

How Patterns In Human Behaviour May Affect The Model.

As this model depends purely on the player-base to determine its data, players 'tilting' and 'throwing' may affect the accuracy of the model's predicted outcome. For example, lakers may stop warding the jungle after being solo killed too many times, and junglers may ignore ganking lanes if they have been flamed by them, players tilted may take worser fights or overextend and throw the lead they have, overall affecting the tempo to which the game's state is currently. These emotional states of players create non-linear, hard-to-predict deviations from standard gameplay. The model may detect gold-lead as a high win-chance, whereas in the actual game players are throwing their lead, and the model has not been able to detect this. If data like this is introduced to the model there may be increased chances of noise, reduced reliability, and forces the model to learn behavioural volatility rather than pure game statistics. League of Legends is also a competitive game featuring a ranked gamemode, this may lead to inconsistency at levels of players, as the model interprets inputted data as purely one-dimensional, lacking the features to differentiate from the gaps that may differ and different ranks, this human inconsistency introduces variance that pure game-state features cannot fully capture. This limits the ceiling of predictive accuracy unless the model incorporates player-level features. Although these are negative, there are many positives to be taken into account when looking at the way players play the game. Through every patch/update/tweak a new meta arises which players tend to often copy, this creates herd behaviour, making specific patterns easier to detect, and way more accurate to predict, improving generalisation. League of Legends also features items that affect how strong a champion is, how tanky they are, and how they influence the tempo of the game, the model at its current state does not take into account the build paths users are taking, only the total gold of the combined team, this leads to scenarios where a team may have exponentially higher gold—which the model detects as an absolute win—they may be building items which are completely useless 'throwing' the game in the later stages, as a proper item on a champion is more likely to increase lane priority against a champion building the wrong statistics on their champion.

Bias in relation to the model.

The model itself uses a single sigmoid curve to differentiate wins from losses. League of Legends outcomes are determined by a multitude of interacting features, as a 3,000 gold lead from the ADC may not mean much at 35 minutes. The model has no way to represent this due to treating all observations equally, regardless of the stage the game is at. The model collapses all determining factors down to a single number (GoldXpAdvantage) and tries to predict state from that alone. This shows underfitting, as it may miss real patterns in the data that are present, but can't comprehend because the model isn't made to express them. The dataset the model is based on is relatively small and narrow due to it being only matches from the EUW region. Just like every other region EUW has its own meta preferences, champion picks, and play-styles that differ from the others. If the model was deployed to another region, or separate ranks, it would likely perform poorly due to the underlying distribution and play-style of matches being different. Relevant in the confusion matrix when precision does not equal recall in the classification report, it reveals a directional bias. If recall for Blue winning is high but precision is low, the model predicts blue winning too relaxed, capturing most real blue wins but also may be incorrectly labelling red wins as blue's. This means that the model is being too overconfident.

The decision boundary sits near 0.035 rather than 0.5 particularly to provide insight to the bias of the model. A boundary shifted toward 0 means the model is treating relatively low GoldXpAdvantage scores as sufficient to predict blue winning, this suggests that the training data had more times where blue won, or the feature actually skews that way in EUW games, but that doesn't apply to the rest of the world.

The model resides around the high-bias zone, consistently making similar mistakes across any new datasets it may need to work off of